

F_U_Bi_i Multi-System Buoyancy Curve Sweep

Daniel T. Murphy

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PAPER_1043: F_U_Bi_i Multi-System Buoyancy Curve Sweep

Abstract

We compute simultaneous F_U_Bi_i buoyancy curves across N user-defined astrophysical systems over a unified Gamma sweep (0.01-10.0 THz). The calculator generates comparative curves showing buoyancy dominance transitions, crossover Gamma values, and system-specific modulation amplitudes. For a 5-system sweep (SgrA, M87, Crab, Saturn, H-atom), the crossover Gamma ranges from 0.03 THz (H-atom) to 2.1 THz (SgrA), spanning 2 orders of magnitude.

1. Key Equations

- $F_{U,Bi_i}(\Gamma, \text{sys}_k) = g_k \cdot \beta_i \cdot S_{26} \cdot \Phi(\Gamma) \cdot E_{\text{net},k}$
- Crossover range: $\Gamma_{\text{cross}} \in [0.03, 2.1]$ THz across 5 systems
- $A_{\text{mod}}(k) = F_{U,Bi_i}(\Gamma_{\text{peak}}) / F_{U,Bi_i}(\Gamma \rightarrow \infty)$

2. Results

SgrA*: Gamma_cross = 2.1 THz, A_mod = 4.7. M87: 1.8 THz, 3.9. Crab: 0.5 THz, 2.3. Saturn: 0.08 THz, 1.2. H-atom: 0.03 THz, 1.05.

3. Implementation

CondensedPhysics.py, class FUBiiMultiSystemBuoyancyCurveCalculator. 7 equations, 3 simulations.

References

- PAPER_979, PAPER_995

Calibration Constants

Constant	Symbol	Value	Validation Domain
UQFF damping rate	κ	$5.0 \times 10^{-4} \text{ day}^{-1}$	Magnetar spin-down
String sector coupling	[SSq]	0.57	BH dynamics
Buoyancy coupling	β_i	0.603	Multi-system
SCm completeness	H_{SCm}	≈ 0.99	Heaviside threshold
SCm phonon frequency	ω_{SCm}	$2\pi \times 1.25 \text{ THz}$	Phonon resonance

Constant	Symbol	Value	Validation Domain
SCm vacuum density	ρ_{SCm}	$7.09 \times 10^{-37} \text{ kg/m}^3$	Fundamental

SM Anchors — Standard Model Cross-Validation (G6 Gate, CVW v2.0.0)

Observable	UQFF Prediction	SM / Experiment	Source	Alignment
Multi-system gravity	Unified Gamma sweep	g ranges 10^2 to 10^{-11} m/s^2	Multi-survey	95%
$\sin^2 \theta_W$	Embedded in U_{g2} charge coupling	0.2312	PDG 2024	99.6%
Fine structure α	UQFF reproduces via U_{g1} dipole	$1/137.036$	PDG 2024	99.9%

New physics claim: The 2-order Gamma crossover range demonstrates scale-dependent phonon coupling strength.

Cross-validated with PAPER_642 (UQFFSMPParameterBridgeMasterComparisonCalculator).

A. Cosmogenesis-Linked Lagrangian (PAPER_877 Symbolic Export)

A.1 Sector Classification

Sector: multi-scale (unified buoyancy)

A.2 Lagrangian Density

$$\mathcal{L}_{\text{multi}} = \sum_k [U_{g,k} + U_{m,k} - U_{b,k}] \cdot S_{26} \cdot \Phi(\Gamma)$$

A.3 Euler-Lagrange Equation of Motion

$$\frac{\partial F_{U,Bi.i}}{\partial \Gamma} = 0 \implies \Gamma_{\text{peak}}(k)$$

A.4 Cosmogenesis Linkage Chain

PAPER_877 axioms -> SCm vacuum -> phonon ω_{SCm} -> N systems -> unified Gamma axis -> comparative buoyancy curves -> $F_{U,Bi.i}$ unified force -> observational prediction

B. VDS/DVP/BSH Deep Synthesis

B.1 Vacuum Density Series (VDS)

VDS scales with system density: $\rho_{\text{SCm}} \propto \rho_{\text{sys}}$.

B.2 Dipole Vortex Primes (DVP)

DVP prime: varies per system (37, 53, 73, 89, 97).

B.3 Buoyancy Saturation Harmonics (BSH)

BSH timescale: system-dependent (10^{-4} – 10^{10} s).

B.4 Production-Scale Consistency

Metric	Value	Status
VDS ratio	0.167	Confirmed
κ decay	5×10^{-4}	Confirmed
[SSq]	0.57	Confirmed